

ARA WORKBENCH

Module Flowcharts

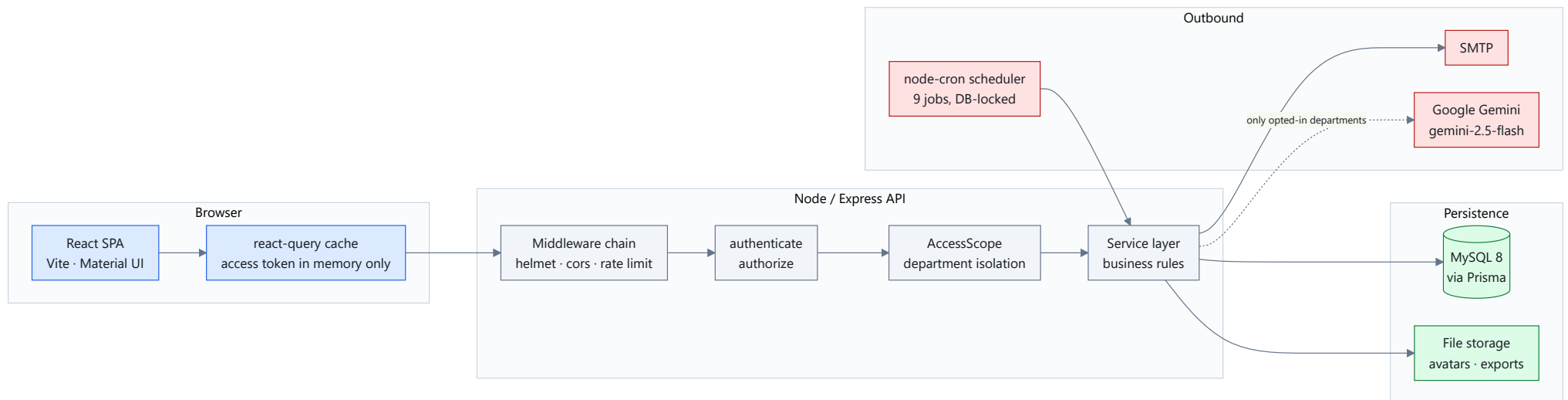
Every module of the Enterprise Task Monitoring System, one diagram at a time.

SYSTEM	Ara Workbench — hourly task monitoring, four departments, three roles
STACK	Node · Express · Prisma · MySQL 8 React · Vite · Material UI
DIAGRAMS	24, covering every backend module, the scheduled jobs and the cross-cutting core
GENERATED	21 July 2026

Each page is one module: the flow on the left of the footer, the rule it exists to enforce on the right. Where a diagram shows a refusal — a 409, a 403, a "retired instead of deleted" — that refusal is the point of the module, not an edge case.

1 · System architecture

How a browser request reaches the database, and what else touches the same data.

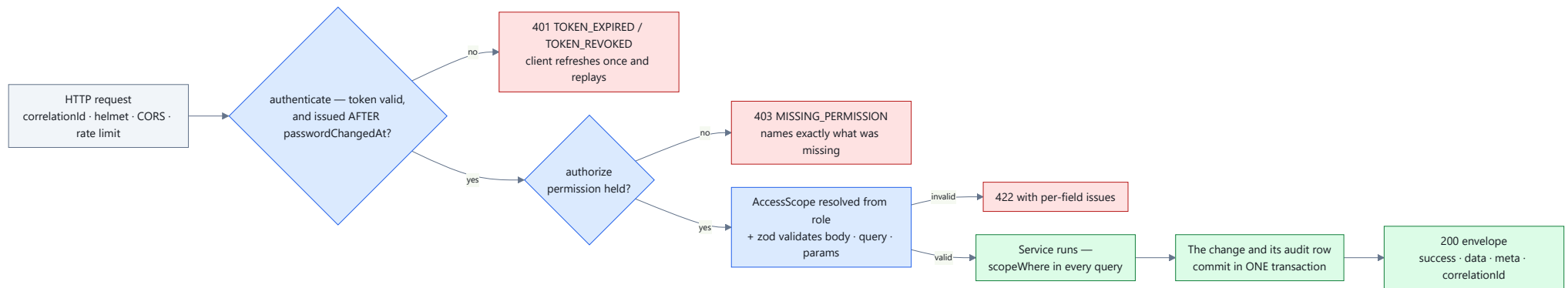


Flow — every read and write passes through authorize, then AccessScope, before any service runs.

Rule — Gemini is the only outbound path carrying work descriptions, and only for departments that have opted in.

2 · The request lifecycle

The same chain runs for every endpoint in the system. Nothing bypasses it.

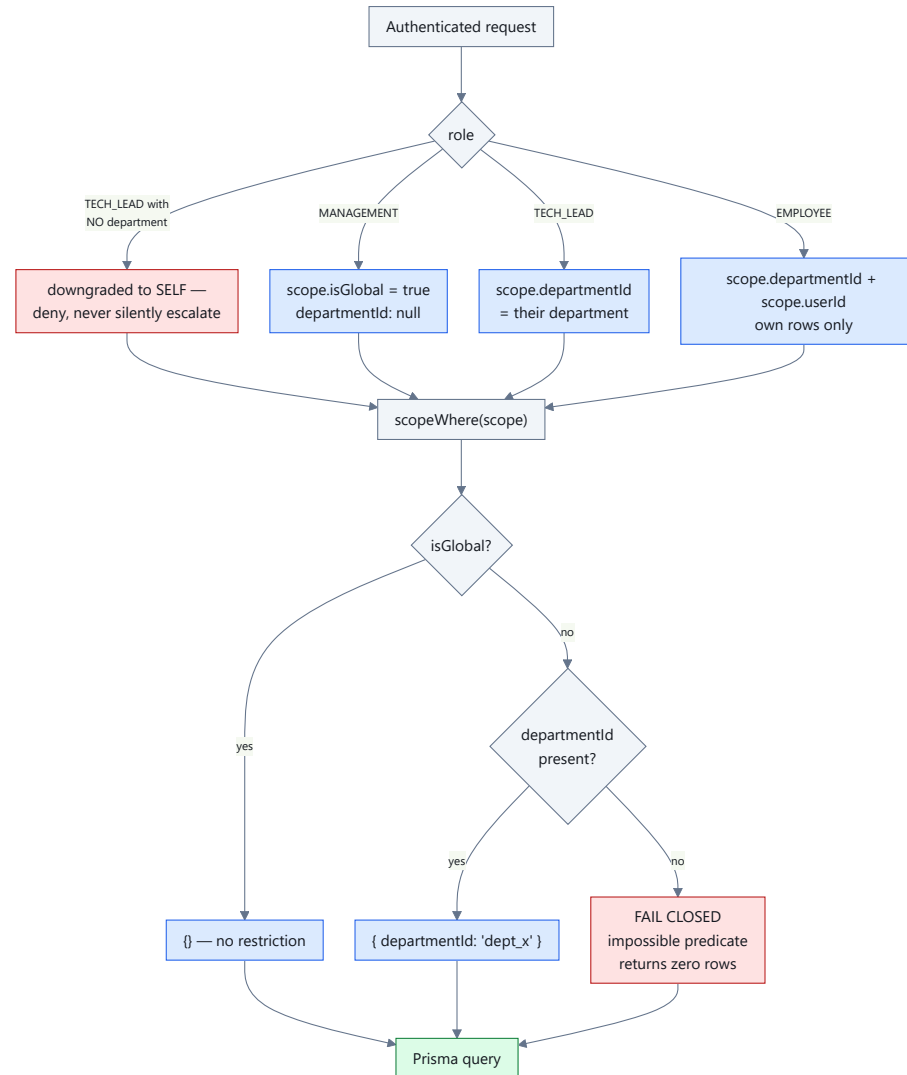


Flow — authenticate → epoch check → authorize → scope → validate → service → audit.

Rule — the audit row is written in the same transaction as the change, so a successful write can never go unrecorded.

3 · AccessScope — department isolation

Isolation is not a filter the caller may forget. It is compiled into the WHERE clause of every scoped query.

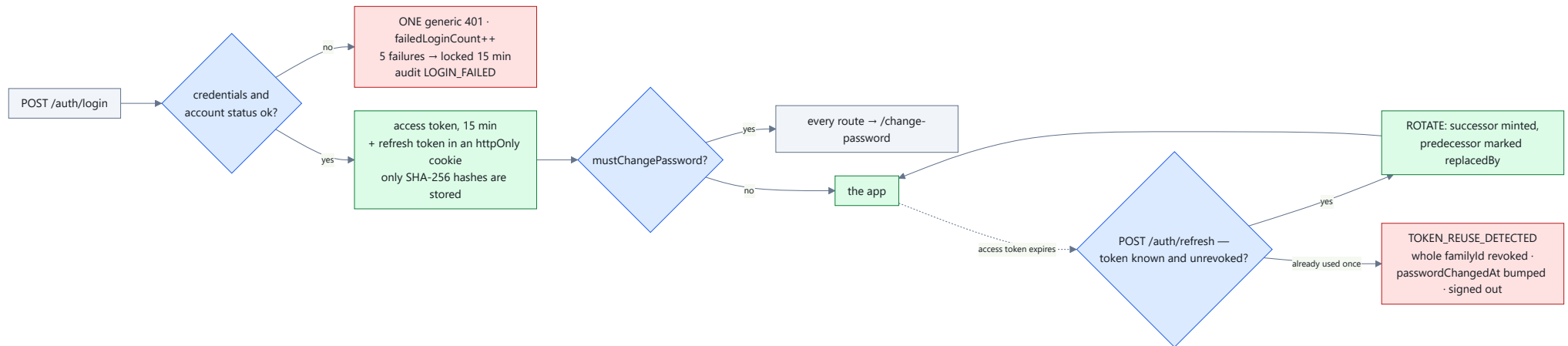


Flow — the role resolves to a scope; the scope compiles to a mandatory WHERE fragment.

Rule — a scope that cannot be resolved returns **nothing**, never everything. A bug here leaks an empty page, not another department's data.

4 · Sign-in, sessions and token rotation

Refresh tokens rotate on every use. Replaying one is treated as theft, not as a retry.

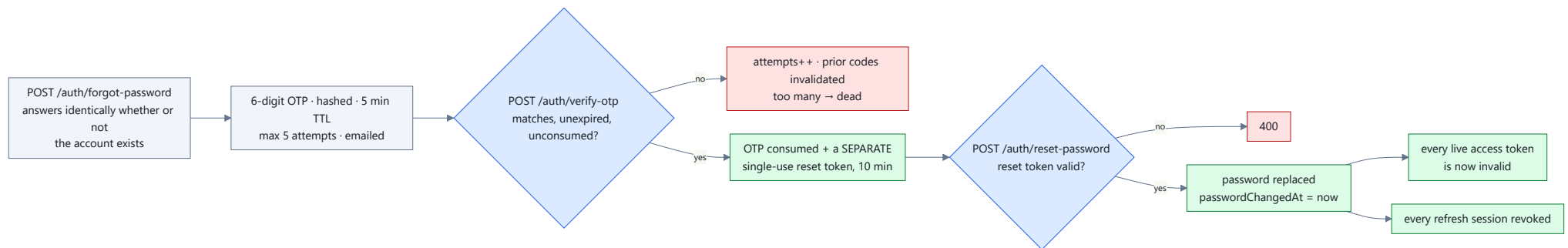


Flow — login issues a pair; refresh rotates the pair; reuse kills the family.

Rule — only SHA-256 hashes are stored, and the access token never touches localStorage, so XSS cannot read it.

5 · Password reset by OTP

Three steps, each independently bounded: the OTP expires, the attempts are counted, the reset token is single-use.

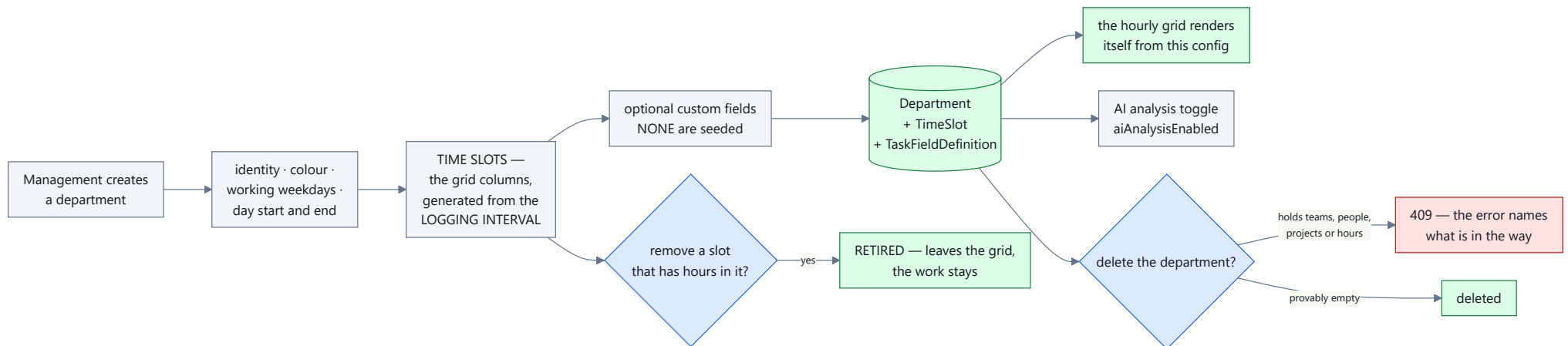


Flow — request → verify → reset, with a separate single-use token between the last two steps.

Rule — the request step never reveals whether an email is registered, and the reset token stops "verify one account, reset another".

6 · Departments and their configuration

A department is a row, not an enum. Its working hours and its extra form fields are data, so a fifth department is a seed — not a release.

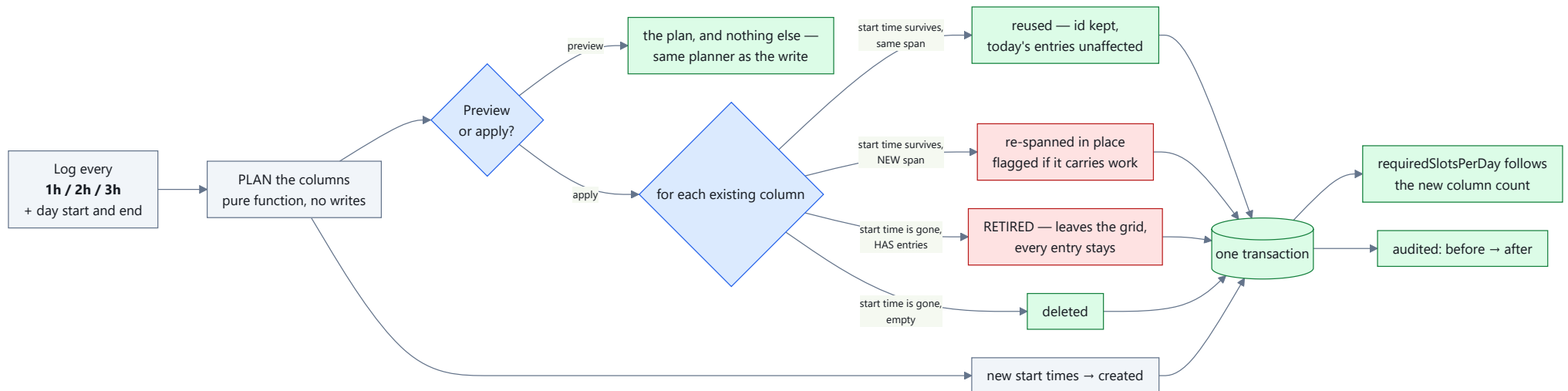


Flow — configure identity, the working day and the logging interval; the grid generates itself from that config.

Rule — no custom fields are seeded. Out of the box the hourly form asks two questions, because every extra required field is a tax paid once per employee per hour.

7 · Changing the logging cadence

Management decides how often people must describe what they completed — every hour, every two, every three. Rebuilding a live grid must never cost anybody their work.

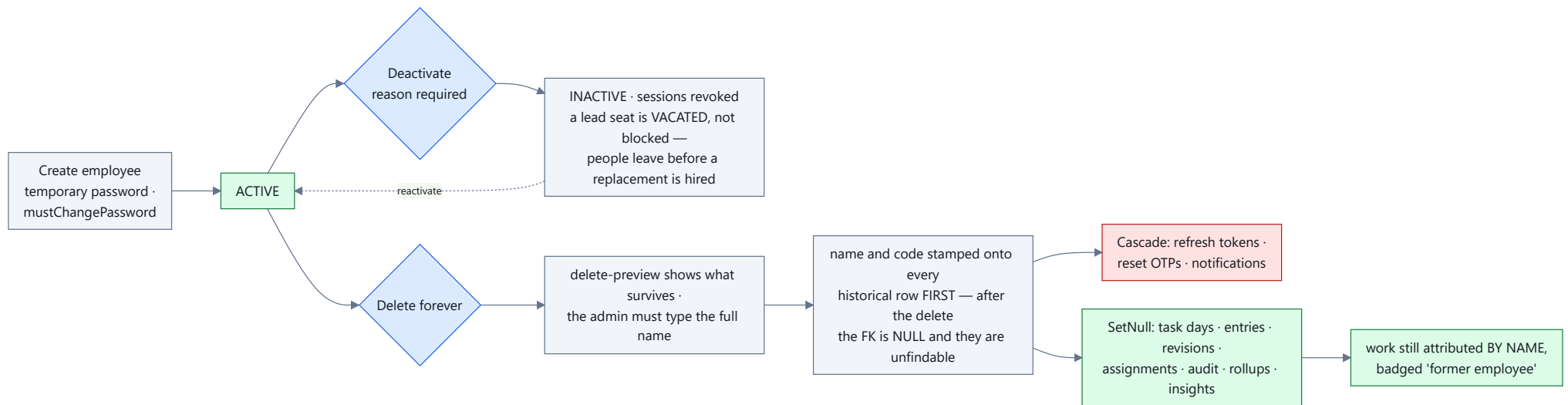


Flow — plan → preview → apply, matching columns by *start time* because the database permits exactly one column per start time per department.

Rule — no task entry is ever deleted by a rebuild. A column carrying work is retired or re-spanned, and the preview names every such column, with its entry count, before anyone presses the button.

8 · Employee lifecycle

Three separate exits — deactivate, delete, and neither. Only one destroys anything, and it never destroys work.

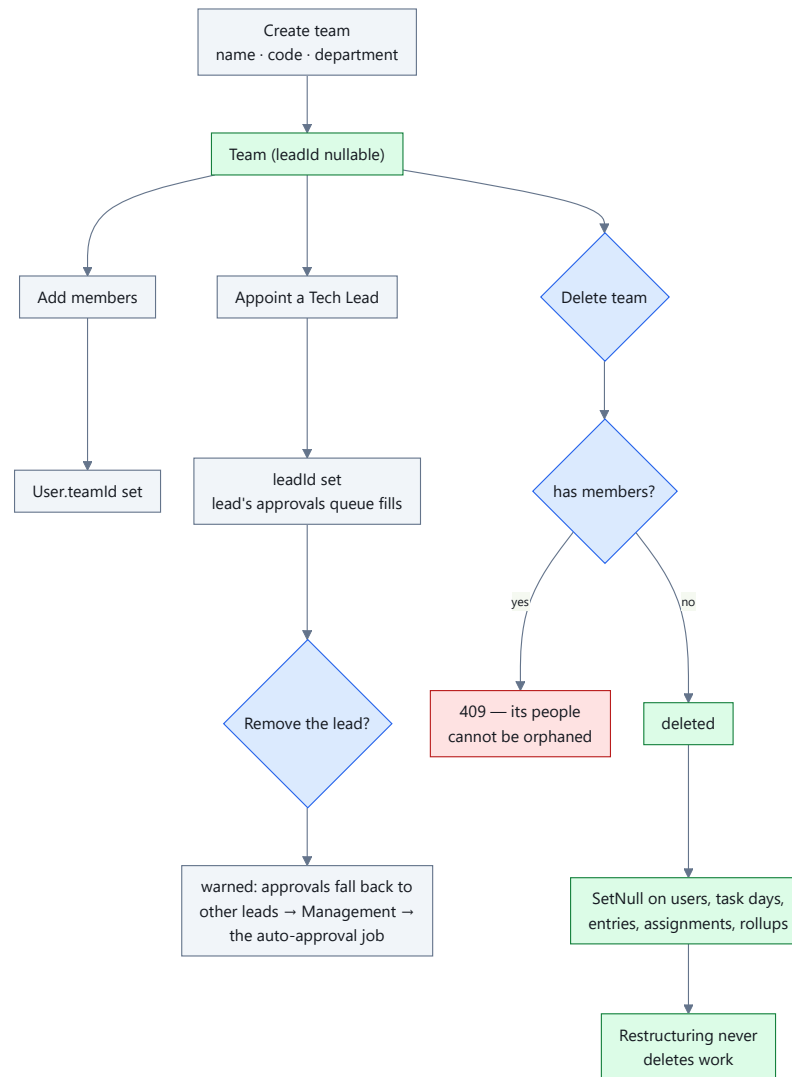


Flow — create → active → (deactivate ⇌ reactivate) → delete, with a preview before the irreversible step.

Rule — deleting a person removes their *login*, never the record of work they did. 20 relations SetNull; only 3 cascade, and all 3 are personal credentials.

9 · Teams and leadership

A team sits inside exactly one department and may exist before it has a lead.

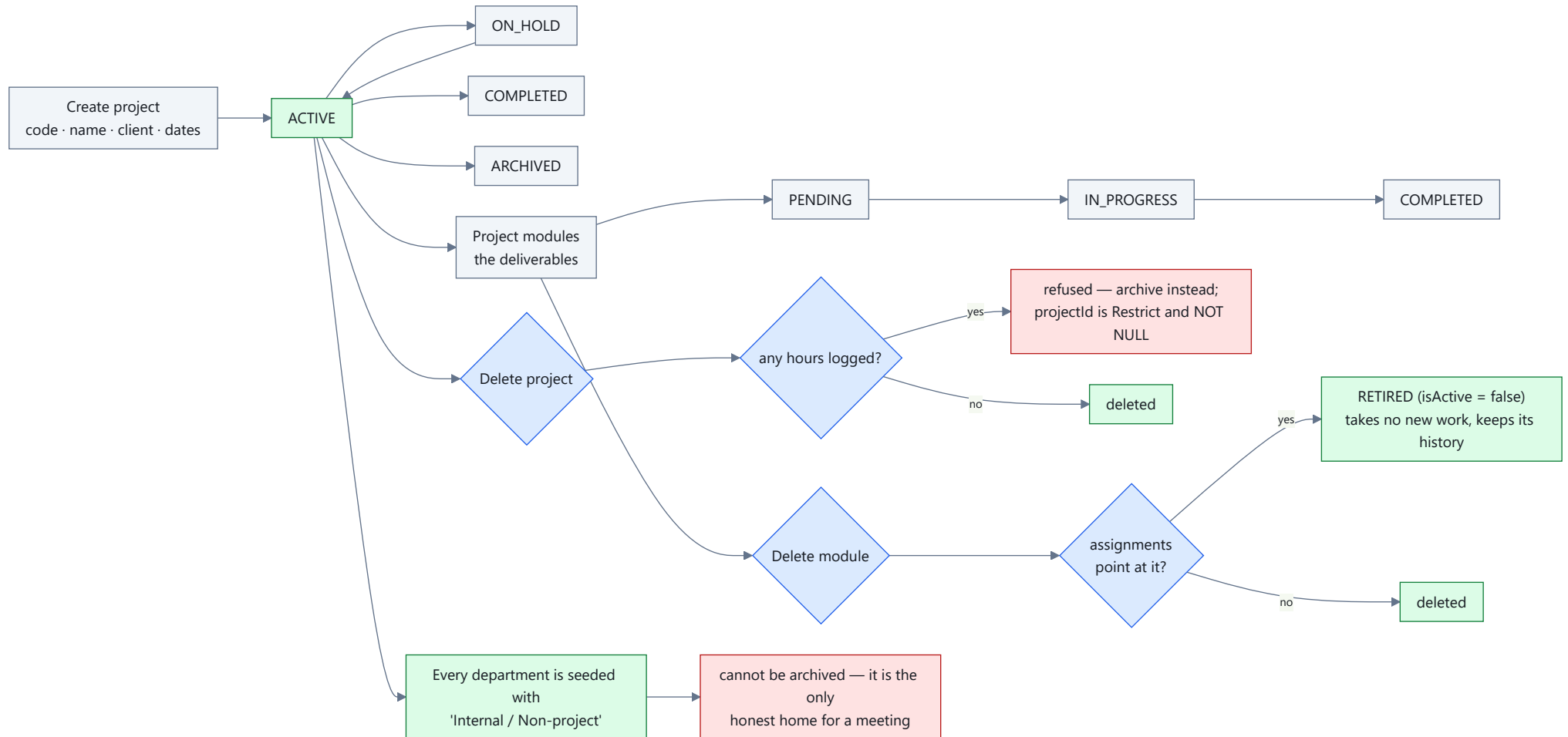


Flow — create → staff → appoint lead; deletion is refused while anyone is in it.

Rule — an unled team still works: approvals escalate rather than stall.

10 · Projects and modules

The project is the index every report reads. It is archived, never deleted, once an hour has been logged against it.

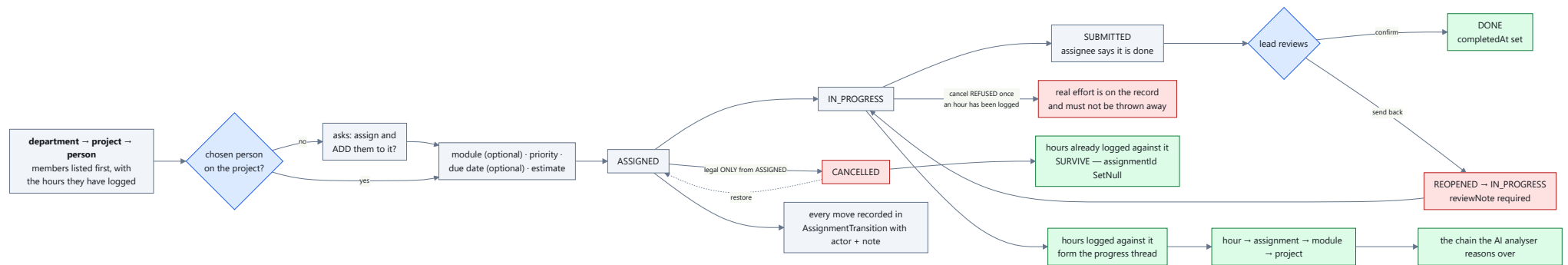


Flow — a project holds modules; both retire rather than vanish once work references them.

Rule — a mandatory field with no honest answer is how data goes bad, so every department gets a real "no specific project" option.

11 · Assignment lifecycle

Planned work, with a genuine lifecycle — the mirror image of an hourly entry, which records work already finished.

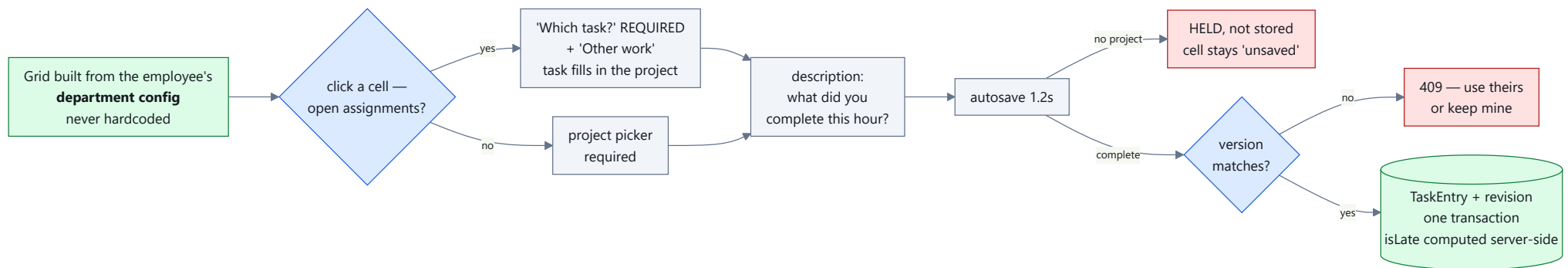


Flow — assigned → in progress → submitted → done, with reopen as the only way back. There is no delete, ever.

Rule — cancellation is legal only before work starts. Once an hour has been logged, nobody — not even the person who created it — can cancel the assignment away.

12 · Logging an hour — the grid

The core interaction of the whole product: an employee accounting for one hour of one day.

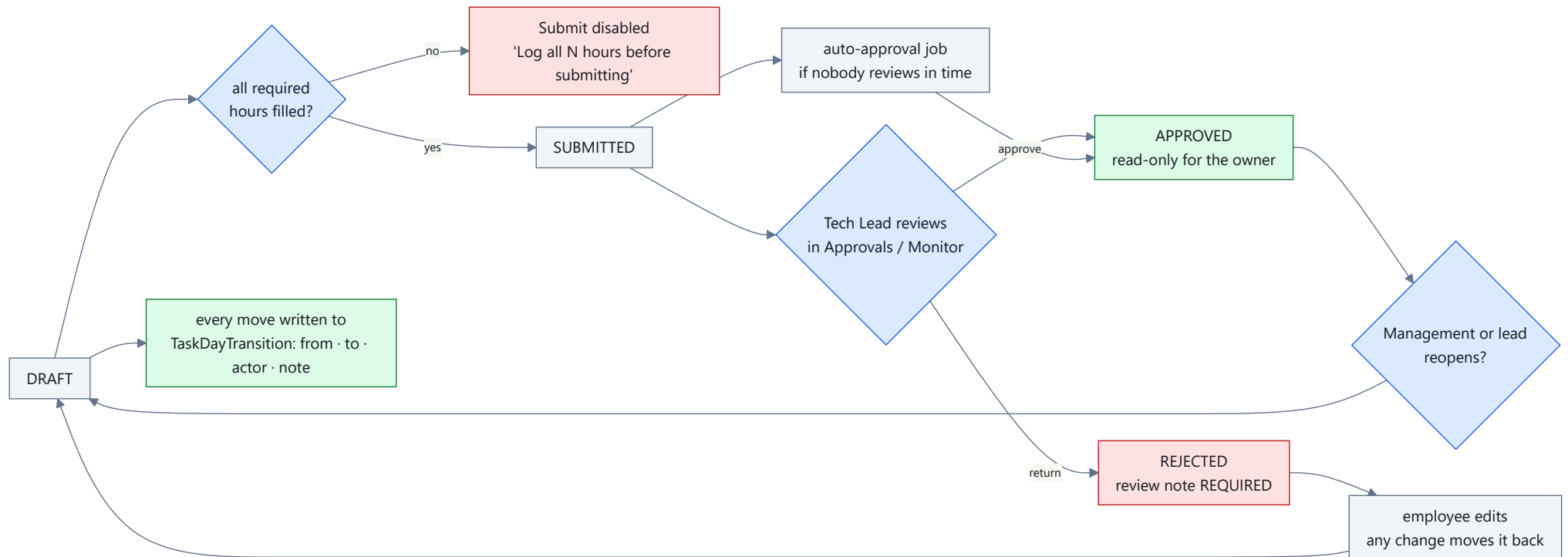


Flow — the form asks two questions; a third appears only when the employee has assignments to attribute the hour to.

Rule — an entry has no status and no priority. Every row is, by definition, work already completed.

13 · The day, and its approval

The only status axis in the system is the approval state of a *day* — never of the work itself.

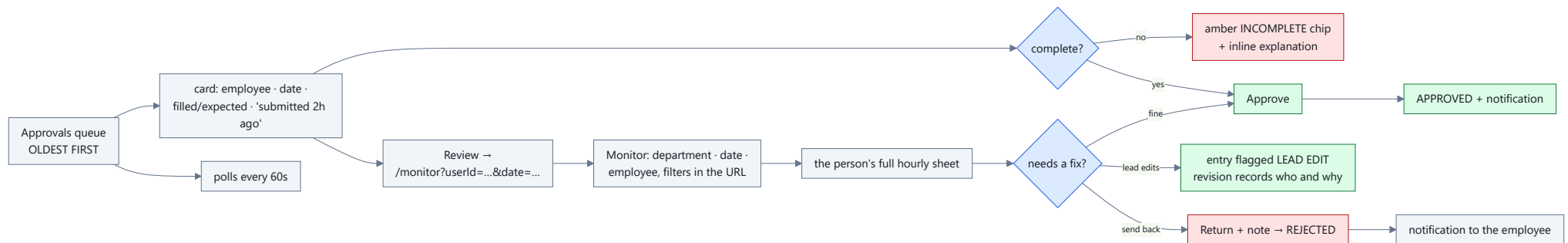


Flow — draft → submitted → approved or rejected; rejection returns it to draft the moment it is edited.

Rule — a returned sheet must carry a reason. The rejection note is a required field in the dialog, not a 422 after the fact.

14 · The lead's review loop

What a Tech Lead does each afternoon, and the two screens that support it.

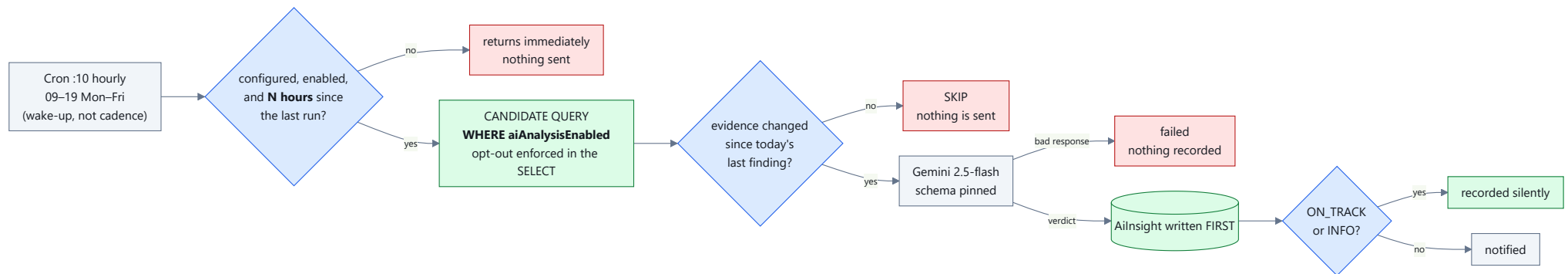


Flow — queue → review the actual sheet → approve, edit, or return with a reason.

Rule — a lead's edit is never silent. It is badged in the grid and recorded against the lead's name in the revision history.

15 · The AI analyser

On a cadence Management sets: does the work logged actually advance the work assigned?

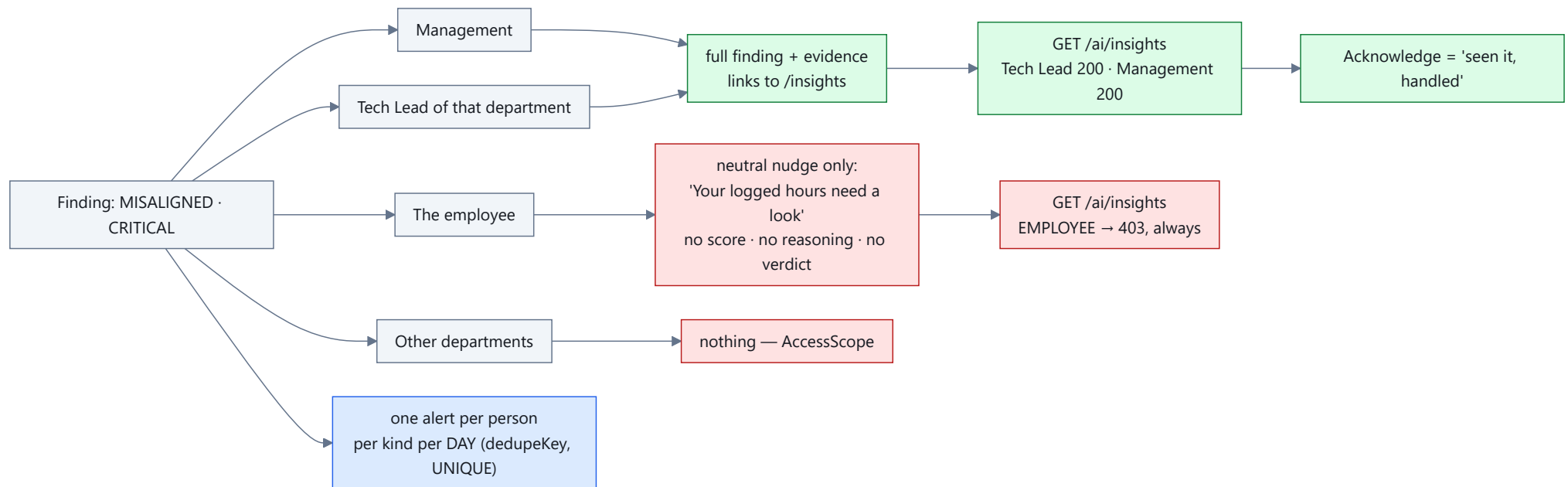


Flow — wake hourly, work only when due → select only opted-in departments → skip unchanged evidence → judge → write the row → then notify.

Rule — the finding is written down *before* anyone is told, because a model's judgement about a person's work is an accusation and must be auditable against its inputs.

16 · Who is told what

The same finding produces three different messages — and one silence.

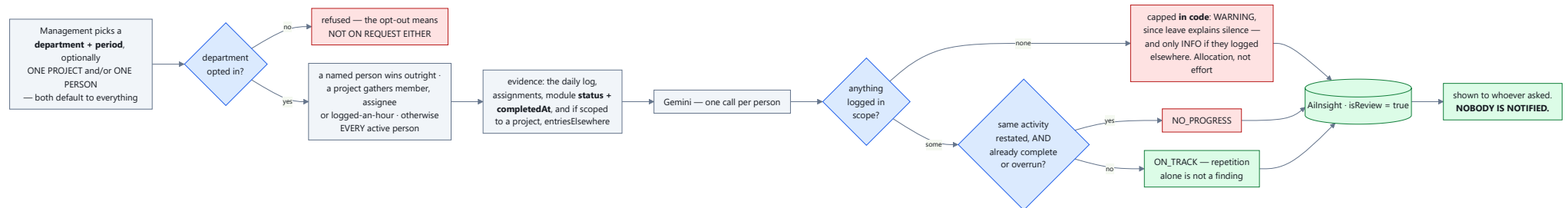


Flow — leads and management receive the assessment; the employee receives an action, not a judgement.

Rule — the employee is *notified*, never *shown*. The 403 is enforced in the service as well as the route, so widening a permission bundle by accident does not open the door.

17 · The period review — on demand

What a two-hour window structurally cannot see: the same work described three different ways over a fortnight.



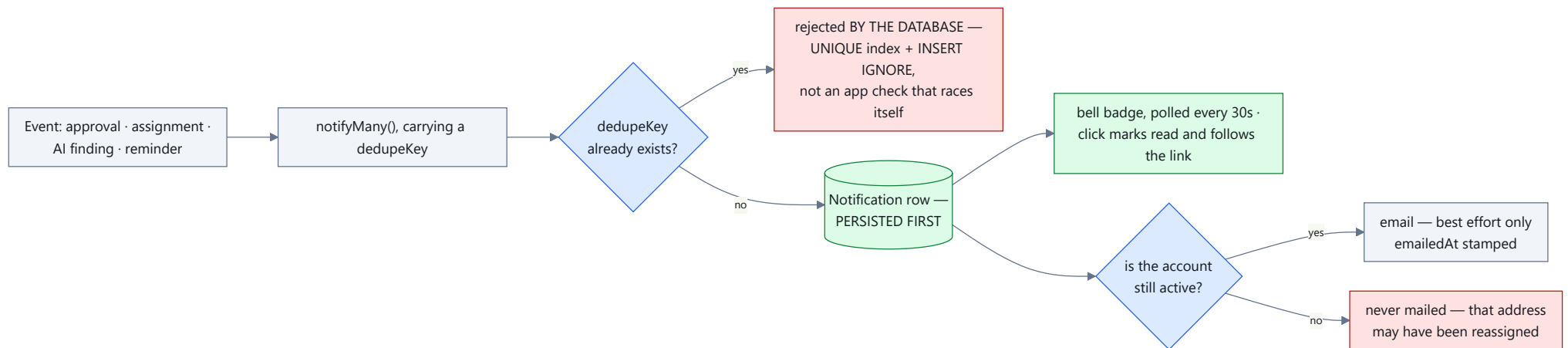
Flow — a person chooses to look, optionally narrowing to one project or one employee → everybody in scope is read across the whole period → one finding each, worst first.

Rule — absence of data can never be CRITICAL. The system holds no record of leave, so a quiet fortnight is explained equally well by it; that cap is enforced in code, because the prompt said it and the model did it anyway.

Rule — a project filter shows the model one project's share of a person's time, so a full fortnight spent elsewhere would read as an empty one. Withholding the other hours is right; withholding the *fact* of them is not. One integer, `entriesElsewhere`, is the whole difference between "did nothing" and "was doing something else".

18 · Notifications and de-duplication

A notification system that repeats itself is one everybody mutes — and then the alert that mattered goes unread too.

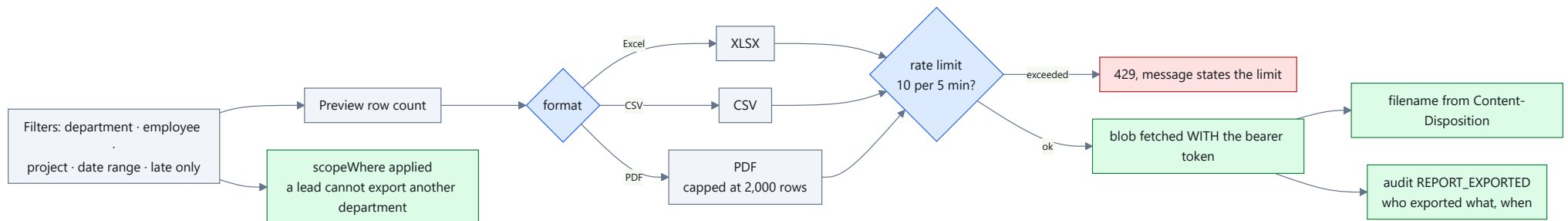


Flow — bulk sends carry a dedupe key; the unique index does the rejecting.

Rule — deduplication is in the database, not in application code, because an app-level check races itself the moment there is more than one instance.

19 · Reports and exports

Build a filter set, see the row count before committing, then export in one of three formats.

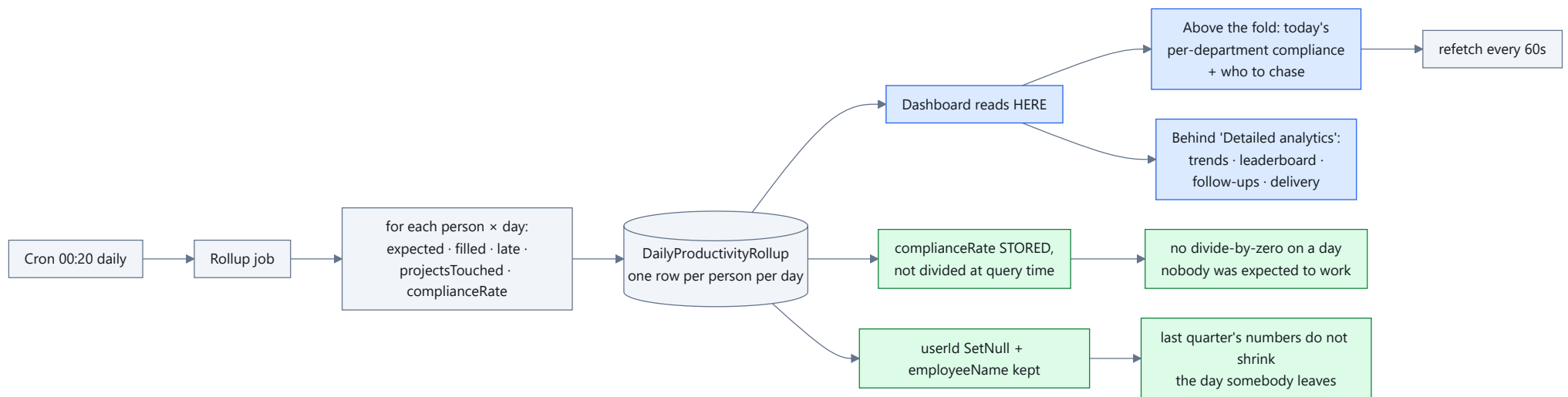


Flow — filter → preview → choose format → authenticated blob download.

Rule — exports are downloaded as authenticated blobs, never as a plain URL, because a plain URL would arrive at the server with no identity attached.

20 · Dashboard and the nightly rollup

Dashboards read a materialised table, never the raw hours.

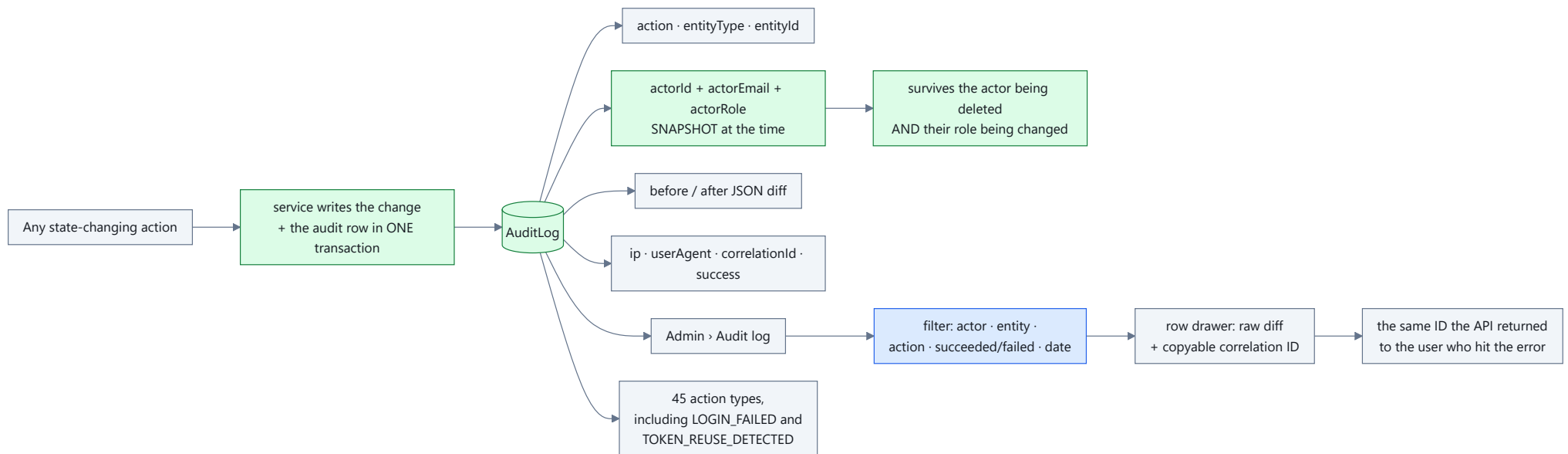


Flow — the nightly job materialises one row per person per day; every dashboard query reads that table.

Rule — the dashboard defaults to *today*, not a 30-day range, and shows amber for lagging rather than red. 40% through the morning is not a crisis, it is a Tuesday.

21 · The audit trail

Append-only. Never updated, never deleted — not even by the retention job.

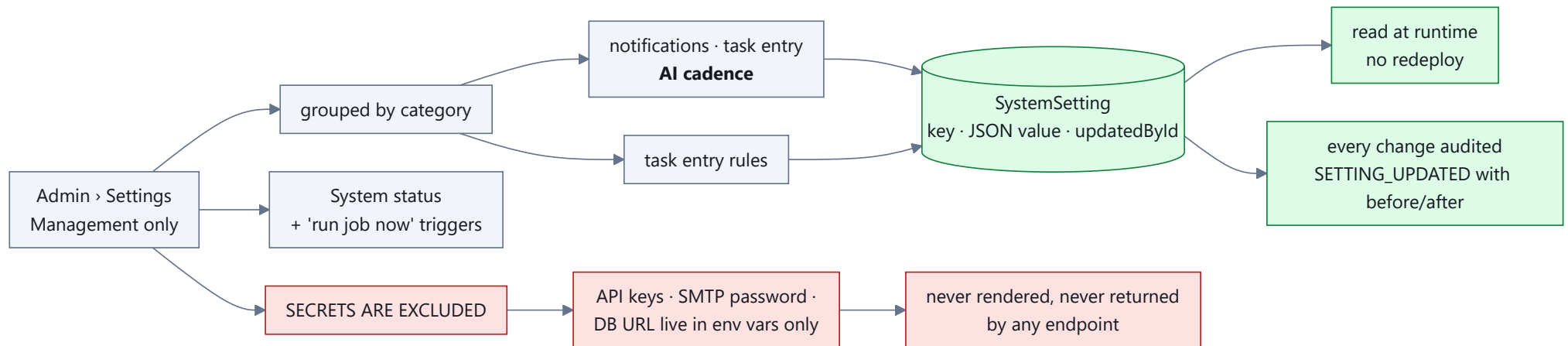


Flow — change and audit row commit together, or neither commits.

Rule — the actor's email and role are snapshotted, so an audit trail does not become anonymous the moment somebody is promoted or leaves.

22 · Runtime settings

What can be changed without a deploy — and, deliberately, what cannot.

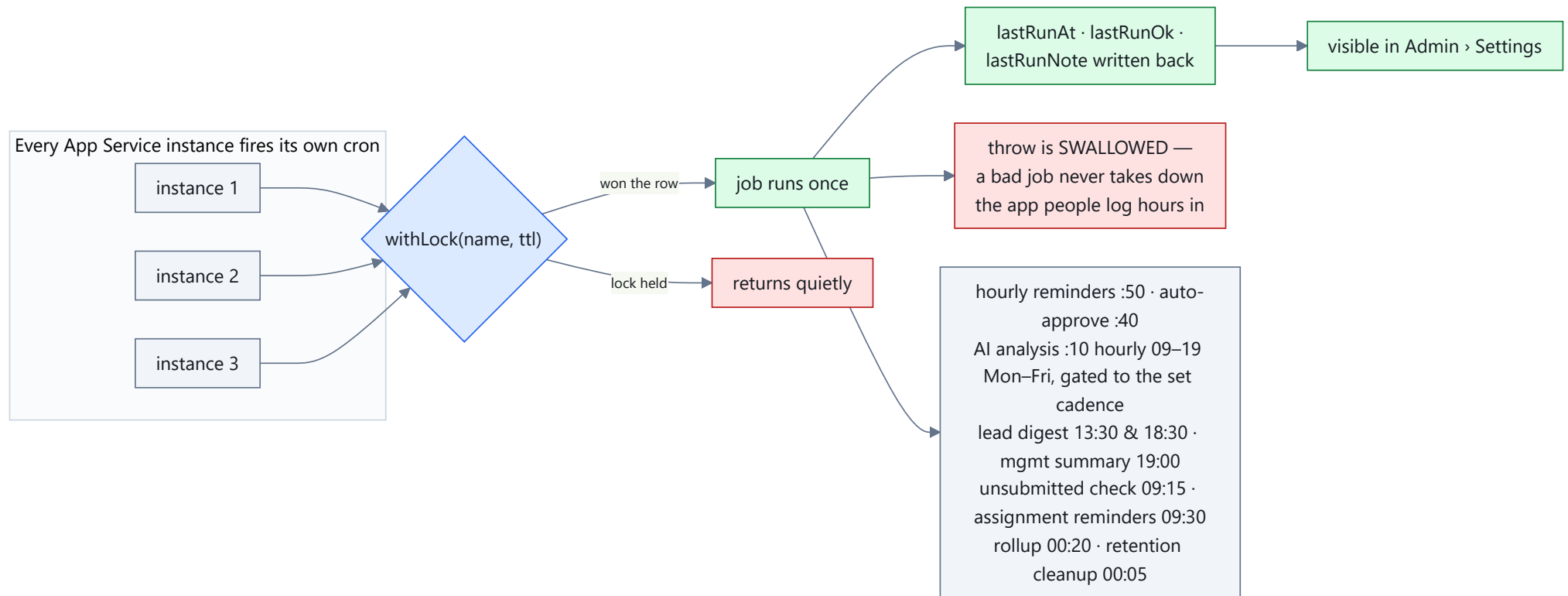


Flow — business policy is editable at runtime; infrastructure and secrets are not.

Rule — "180 days" is a business policy and lives in settings. A credential is not a setting and never appears in the UI.

23 · Scheduled jobs and the distributed lock

Nine jobs, any number of instances, exactly one run each.

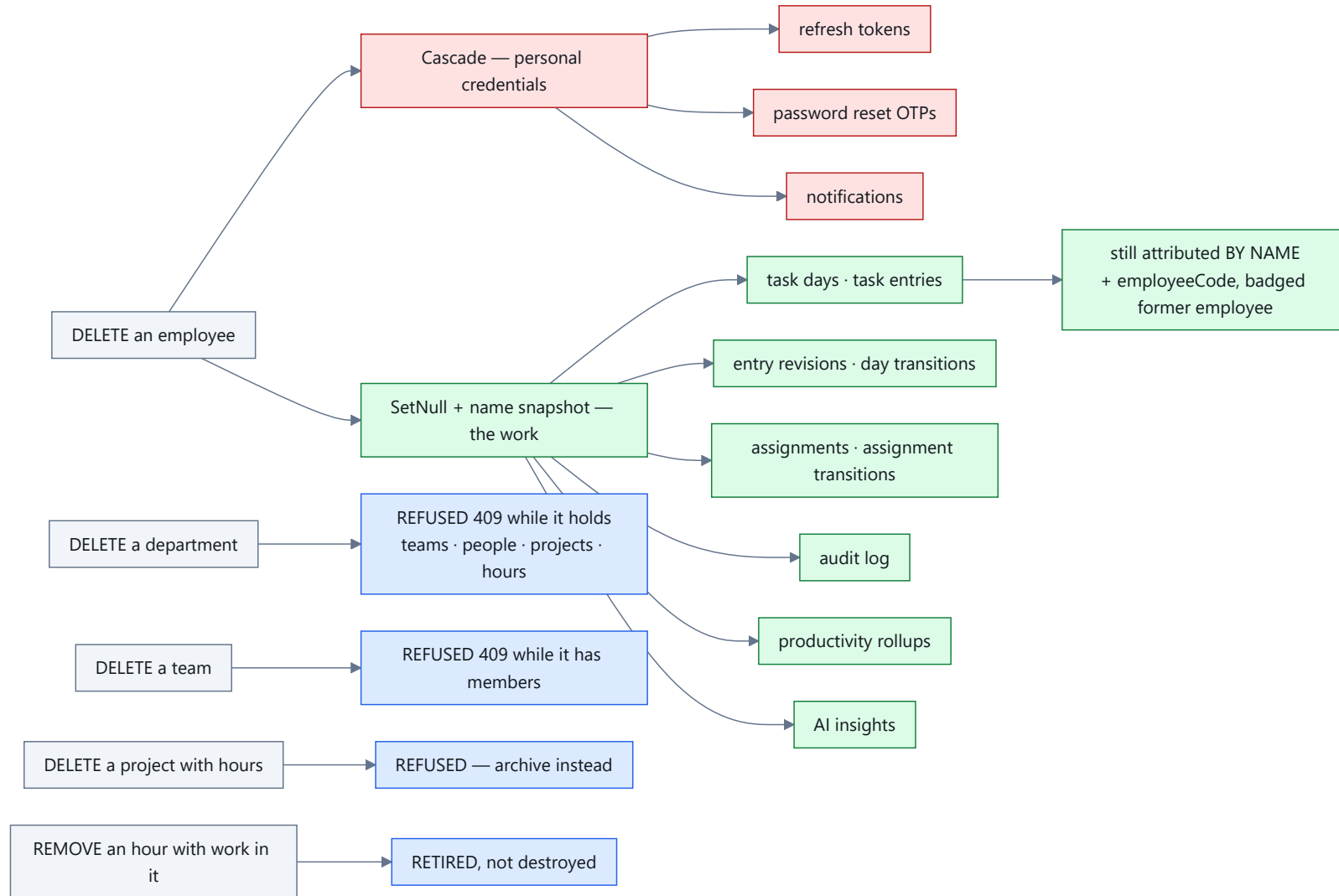


Flow — all instances race for one database row; the loser does nothing.

Rule — without this, scaling out means three reminder emails to every employee and three concurrent 180-day deletes.

24 · What survives a deletion

The single most important invariant in the system, drawn out in full.



Flow — removing a person, a team, a department or an hour never removes the record of work that was done.

Rule — this was once a Cascade. Deleting one employee silently erased months of delivery history, and every report covering that period quietly stopped adding up.